ZHENGYU CHEN

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EXPERIENCE

Palatial - Agent Engineer *Shanghai, China* *Jun 2026 - Present*

- **Agent & Durable LLM Runtime:** Owned end-to-end design and productionization of an AI Agent on **Cloudflare Workers / Workflows / Durable Objects / R2**—multi-step tool orchestration with checkpointed/resumable turns, retry and recovery under failure/redeploy, and durable session state. Integrated **Vercel SDK + Gemini Flash** for intent classification, long-horizon **context management**, and reliable agent–tool contracts across releases.
- **Scene Reconstruction Research & Multi-Agent 3D Assets:** Led literature review and paper-driven R&D on **vision / scene reconstruction**, and implemented reconstruction-to-sim-ready-asset pipeline; built the **Blender Agent** for object / **3D asset** creation on the **Fable** and **Astra** harness; and orchestrated multi-agent workflows with consistent **context management** across agents.

ByteDance - Agent Engineer *Shanghai, China* *Jan 2026 - Jun 2026*

- **Multi-Agent Architecture & Skill Discovery**: Helped evolve the agent system from a single-agent execution model into a **multi-agent architecture**, enabling more reliable execution of complex, long-horizon tasks. Refined the **ReAct loop** and improved skill loading with a service-discovery-inspired mechanism for registering, discovering, and invoking reusable agent skills.
- **Tool Integration & Rich Output Generation**: Expanded the agent system's tool ecosystem by integrating additional external tools, strengthening the LLM's ability to perform tool calls, execute actions, and generate richer types of outputs. Optimized long-session context management through pointer-based replacement of large tool results and threshold-based memory compaction.

Connexion Ltd - Software Engineer *Melbourne, Australia* *Aug 2023 - Nov 2025*

- **High-Concurrency Data Architecture & Computation Offloading**: Engineered a robust storage architecture to tackle multi-tenant write bottlenecks. **Designed business-aware sharding strategies** for DynamoDB Partition Keys to ensure uniform write distribution and eliminate throttling. Architected a **Redis + DAX multi-tier caching system** with **Redis computation offloading** to decouple complex aggregations from the database, reducing P99 write latency to **180ms**.
- **Zero-Downtime Migration & Event-Driven Scaling**: Spearheaded the **zero-downtime migration** of 3M+ mission-critical records from MongoDB to DynamoDB. Established **CDC real-time sync pipelines** with **shadow write verification** to guarantee 99.99% consistency. Integrated **AWS SQS + KEDA** to orchestrate event-driven autoscaling based on **queue backlog**, ensuring system stability during traffic bursts.

Amazon - Software Engineer *Beijing, China* *Feb 2022 - Mar 2023*

- **Data Proxy Platform Architecture & Optimization**: Architected the data access layer for Retail services. Mitigated connection storms by integrating **RDS Proxy** (reducing connections by **90%**) and transitioning protocols to **gRPC + Protobuf** (slashing P99 latency by **60%**). Engineered a **Binlog CDC-based Cache-Aside** system with **"Read-After-Write" Smart Routing** to resolve data inconsistencies caused by replica lag, successfully supporting million-level QPS during Black Friday.

Tencent - Software Engineer *Guangdong, China* *Jan 2020 - Feb 2022*

- **Gameplay System Development**: Spearheaded the design of core gameplay systems, constructing a **guild-based shared resource pool** to efficiently handle resource allocation and reclamation under high concurrency. Engineered robust data structures for complex operational events, ensuring high system stability and quality delivery.
- **Matchmaking Engine Optimization**: Identified performance bottlenecks via profiling and re-architected the engine using a **Global Object Pool + Smart Pointer Zero-Copy** design. Transformed object queues into

pointer passing mechanisms to eliminate memory allocation overhead and fragmentation. Boosted QPS from 28,000 to 38,000 and reduced **P95 latency to 180ms**.

- **LSTM-Based Predictive Autoscaling**: Developed a predictive scaling system to address native HPA lag. Trained an **LSTM time-series model** on massive historical traffic data and implemented a **K8S Custom Metrics Adapter** to trigger **proactive scaling** 30 minutes in advance, increasing cluster resource utilization from 45% to 73% while maintaining strict SLAs.

EDUCATION

The University of Melbourne *Melbourne, Australia* *Jul 2018 - Dec 2019*
Master of Information Technology
East China Normal University *Shanghai, China* *Sep 2013 - Jun 2017*
Bachelor of Software Engineering

SKILLS

- **Languages**: **C++**, Java, Python, TypeScript, SQL
- **Cloud & Infrastructure**: **AWS** (EKS, Lambda, SQS, RDS), Kubernetes, Docker
- **Data & Storage**: **DynamoDB** (Design & Tuning), Redis Cluster, MongoDB, MySQL (Binlog/CDC)
- **Frameworks & Tools**: gRPC, Protobuf, Prometheus, Grafana, CI/CD, Git
- **Certifications**: **AWS Certified Solutions Architect - Professional**